Date 21/09/2026 Statics

Measure Of Central Tendency:-

Objectives:
→ Summarize data
→ comparison two or more data
→ Provide base for calculating other statistical measures.

Measure Of Central Tendency:
→ Arithemic Mean
→ Geometric Mean
→ Harmonic Mean
} Mathematical averages
→ Median
→ Mode

Criteria or Properties

→ Easily calculated & understandable.
→ Based on all values.
→ Does not effected by extreme values.
→ It can be used further statistical analysis
→ Sampling stability.

How to handle extreme values?
Negotiate if not then remove.

Arithemic Mean:

$$\mu = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{N}$$

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Geometric Mean:

$$G.M = (x_1 \cdot x_2 \cdot \cdots x_n)^{1/N}$$

OR

$$G.M = \text{Antilog}\left(\frac{\log x_1 + \log x_2 + \cdots + \log x_n}{N}\right)$$

Harmonic Mean:

$$H.M = \frac{N}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$$

Median:

does not effect by extreme value.

Mode:

Most frequent/ value in the data.
repeated

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Mid-Term Marks:

S.NO	Male	Femal
01	18	20
02	15	7
03	22	24
04	24	24
05	16	20
06	21	
07	22	
08	18	
09	20	
10	24	
11	21	
12	23	
13	22	
14	19	

Find Arithemic, Geometric, Harmonic Mean
Median, Mode.

Mode : 24

Arithemtic Mean: $20+7+24+24+20 \,/\, 5$

$= 95 \,/\, 5 = 19$

$G.M = (20 \times 7 \times 24 \times 24 \times 20)^{1/5}$

$= (1612,800)^{1/5} = 17.43$

$H.M = \frac{1}{20} + \frac{1}{7} + \frac{1}{24} + \frac{1}{24} + \frac{1}{20}$

$= 0.05 + 0.14286 + 0.04167 + 0.04167 + 0.05$

$= 0.3262$

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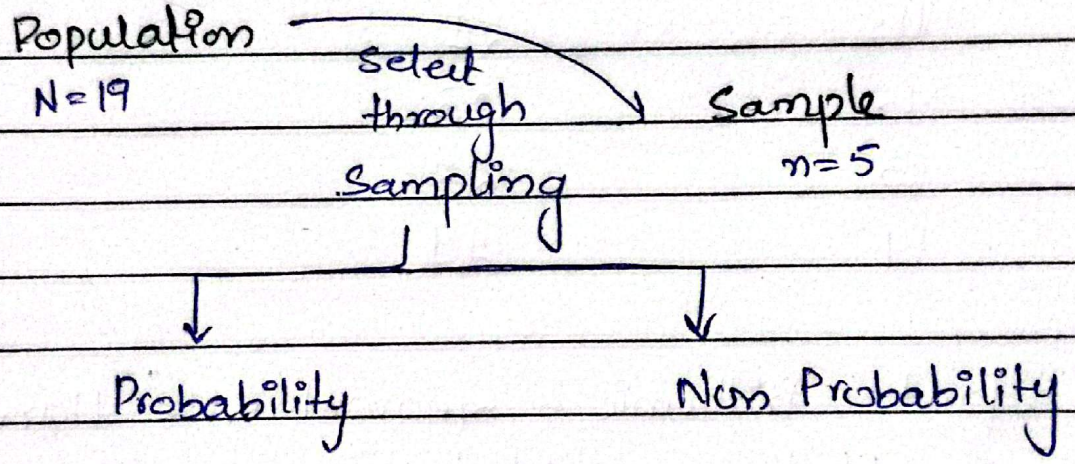
$$H.M = \frac{5}{0.3262} \approx 15.33$$

Median: 7, 20, 20, 24, 94

$n = 5 \rightarrow$ odd

$$\text{Median} = \left(\frac{5+1}{2}\right) \Rightarrow 3\text{rd term}$$

Median = 20



⇒ Absolute Measure
- → Range
- → Mean Deviation
- → ~~Var~~ Standard "

⇒ Relative Measure.
- → coefficient of Range
- → " " Mean
- → coefficient of variation

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Scale :

→ Nominal = , ≠
→ ordinal = , ≠ , > , <
→ Interval = , ≠ , > , < , + , -
→ Ratio = , ≠ , > , < , + , - , ÷ , x

Data → ordinal scale
Sample → simple random sample

S.No	X	X-76	|x-67|	$(x-76)^2$
	75	8	8	64
	71	4	4	16
	81	14	14	196
	57	-10	10	100
	51	-16	16	256
		0	52	632

Range = 81-51 = 30
Mean deviation: 52/5 = 10.2

$$S = \sqrt{\frac{632}{5}} = 11.24$$

coefficient of Range = 81-51 / 81+51 = 131.37
Coefficient of mean deviation = 10.2/76 = 0.134
C.V = 11.24 / 76 × 100 = 15

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Exercise :

25, 41, 27, 32, 43, 66, 35, 31, 15, 5, 34, 26, 32, 38, 16, 30, 38, 30, 20, 21

Find Q1, Q3,

IQR = Q3 − Q1

Lower Limit = Q1 − 1.5 IQR

Upper Limit = Q3 + 1.5 IQR

Ascending order

5
15
16
20
21
25
26
27
30
30
31
32
32, 32, 34
35
38, 38
41
43
66

L.L =

→ Q2 = 30+31/2 ⇒ 30.5

→ Q1 = 21+25/2 ⇒ 46/2 ⇒ 23

→ Q3 = 35+38/2 ⇒ 73/2 ⇒ 36.5

→ IQR = 36.5 − 23 ⇒ 13.5

→ L.L = 23 − (1.5 × 13.5)

23 − 20.25 ⇒ 2.75

→ U.L = 36.5 + (1.5 × 13.5)

= 36.5 + 20.25 ⇒ 56.75

→ Potential outliers is one which is 66.

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Assignment 02

SNo, Subj, T.M, GP, LG

your result → take as a dataset

Simple Bar chart
Component Pie chart
Multiple Bar chart

Result Table:

Mearsures	T.M	Grade	Point
A.M			
G.M			
H.M			
Mean			
Median			
Mode			

⇒ Graphs + discussion
⇒ Q1 + Q2 + Q3
⇒ IQR + Outliers
⇒ Today's lec

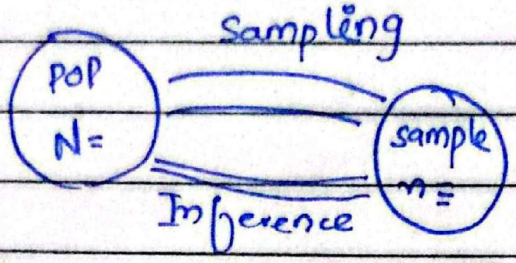
⇒ Find Best Measure?

⇒ GPA ⇒ How to find it?
⇒ CGPA ⇒ How to find it?

Date: 05/Oct/2026 Statistic

Probability Theory
↓ models / solves the problem
uncertainty → Randomness

Descriptive Statistics:



sample	μ	$\bar{X}$
s.varience	σ^2	σ^2

Probability / Inferential Stats: → use to compute data for population on basis of sample.

Types of Statistical Population:
- → Real Population
- → Hypothectial Population

Real Population	Hypothectial Population
→ Past / Present data	→ future assumption
→ physically exist	→ theoritical

Cause of Uncertainty:

Randomness	Non-Randomness
↓	↓
unpredictable, not in our control	Analyzeable

→ Deterministic Experiment

Result's predictable

→ Random / Probabilistic Experiment

Unpredictable Result even under large number of repeatition / under similar condition.

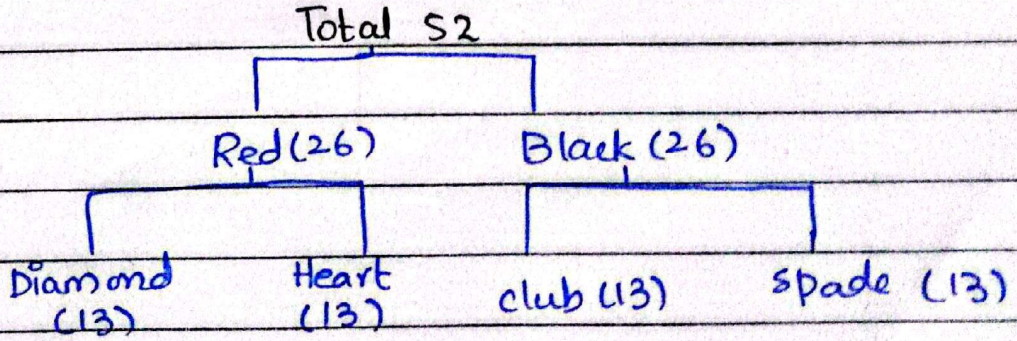
→ sample Space

Set of all possible outcomes of random experiment - (written in set form { }).

	Outcomes	Sample Space
Mth262	A, B, C, D, F	$S = \{A, B, C, D, F\}$
CSC 301	A, B, C, D, F	$S = \{A, B, C, D, F\}$

$S = \{AA, AB, AC, AD, AF, BA, BB, BC, BD, BF, CA, CB, CC, CD, CF, DA, DB, DC, DD, DF, FA, FB, FC, FD, FF\}$

Cards :



Ace - 1

Face (K, Q, Y) = 3

Number 2, 3, 4, 5, 6, 7, 8, 9, 10

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Event & its Types:

Subset of sample space.

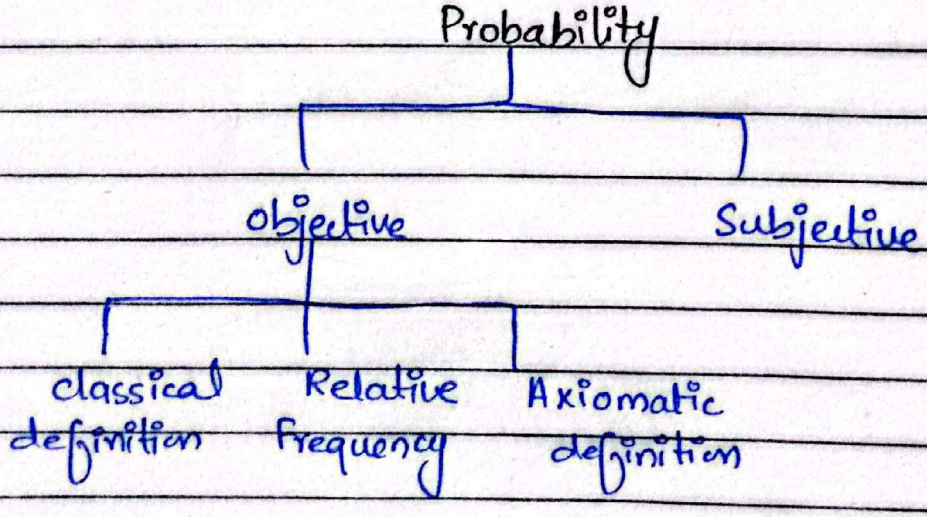
- NULL Event → no outcome (impossible) → {Φ}
- SURE Event → sample space
- Simple Event → 1 outcome
- Compound Event → 2 or more outcomes
- Mutually Exclusive Events → intersection {B∩D} = S (2 events not occur at a time)
- Exhaustive Events → Union {B∪D} = S
- Equally Likely Events → 50% 50% chances

$B = \{F\}$

$D = \{A, B, F\}$

$B \cap D = \{F\}$

Definition Of Probability



Classical or Prior definition:

$$\text{classical (Event)} = \frac{\text{Favourable outcomes}}{\text{Possible outcomes}}$$

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DEFINITION OF PROBABILITY..

Classical or Prior definition

↓

$$P(A) = \frac{n(A)}{n(S)}$$

Mutually exclusive events.

Given $S = \{HH, HT, TH, TT\}$

Find $P(\text{one Head, one tail})$

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{4} = 0.5 = 50\%$$

$$A = \{HT, TH\}$$

⇒ at least one Head

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{4} = 0.75 = 75\%$$

$$A = \{HH, HT, TH\}$$

↳ others

Q# of getting a 3 or a diamond (13)

$14 - 1 = 13 \Rightarrow 13 + 3 = 16$

$$P(d) = \frac{16}{52} = \frac{4}{13}$$

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Q# {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}

① at least 2 head

$$P(h) = \frac{4}{8} = 0.5 = 50\%$$

② at most 2 head

$$P(h) = \frac{7}{8} = 0.875 = 87.5\%$$

③ exactly 2 head

$$P(h) = \frac{3}{8} = 0.375 \Rightarrow 37.5\%$$

L.G	f	r.f
A	2	2/23 =
B	10	10/23 =
C	8	8/23 =
D	2	2/23 =
F	1	1/23 =
	23	

Q# Blood Types:

O	21
A	22
B	5
AB	2
	50

⇒ Quiz from Ex 4.3

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① person having O-Blood Type:

$$P(O) = \frac{\text{fregency of O-Blood Type}}{\text{Total number of people}}$$

$$= \frac{21}{50} = 0.42 = 42\%$$

② person bas type A or type B

Exercise #4.3

4.66 Playing Cards:

R = event a red card is chosen

Probability :

$$P(R) = \frac{26}{52} = 0.5 \Rightarrow 50\%$$

4.67 Poker chips

12 poker chips, 3 red, 4 white, 5 blue

Total = 12

B = event the chip selected is blue

$$P(B) = \frac{5}{12} \approx 0.4167$$

4.68 A lottery

You hold 20 out of 500 tickets.

G = event → win grand prize.

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$$P(G) = \frac{20}{500} = \frac{1}{25} \Rightarrow 0.04$$

4.69 Ages of Senators:

A = Under 50 = 12
B = 50 - 59 = 33
C = 60 - 69 = 32
D = 70 - 79 = 18 E = 80 or above = 5
Total seats = 100

(a) Senater under 70

(A) (B) (C)

frequency = 12 + 33 + 32 = 77

$$P(S) = \frac{77}{100} = 0.77 \Rightarrow 77\%$$

(b) Event express S in terms of events A, B and C

S = A or B or C
(or S = A ∪ B ∪ C)

(c) Determine P(A), P(B), P(C)

P(A) = 12/100 = 0.12 = 12%
P(B) = 33/100 = 0.33 = 33%
P(C) 32/100 = 0.32 = 32%

(d) compute P(S) using addition rule:

P(S) = P(A) + P(B) + P(C)
P(S) = 0.12 + 0.33 + 0.32 = 0.77

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4.70 Sales Tax Receipts

Frequency Table. ...

Total no. of corporation (N) = 25 + 23 + 21 + 11 + 34 + 44 + 17 + 32 = 207

(a) find P(R) where R = receipt < 50,000 ≥ 500,000

→ count for < 50,000 : 23 + 25 = 48
→ count for ≥ 500,000 : 17 + 32 = 49
→ Total frequency = 48 + 49 = 97

$$\rightarrow P(R) = \frac{97}{207} \approx 0.4686$$

(b) R = A or B or C or D
(R = A ∪ B ∪ C ∪ D)

(c) P(A) = 25/207 = 0.1208
P(B) = 23/207 = 0.111
P(C) = 17/207 = 0.0821
P(D) = 32/207 = 0.1546

(d) P(R) = P(A) + P(B) + P(C) + P(D)
P(R) ≈ 0.4686

4.71 Twelfth-Grade Smokers

Percentage table :

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(a) Smoked

P(A) = 73.3%

P(Smoked = 100% - P(A)

100 - 73.3 = 26.6%

= 0.267

(b) Smoked at least one:

Sum of C, D, E, F, G ⇒ 16.9%

0.169

(c) Smoked blw 6 - 34 per day:

Sum of D, E, F = 8.8% = 0.088

4.72 Home Internet Access

Percentage Table ...-

(a) Under 75,000

P(A or B) = 47.5% + 19.9%

= 67.4% = 0.674

(b) 50,000 or above

P(B or C or D) = 19.9% + 24.7% + 7.9%

= 52.5% ⇒ 0.525

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(c) Blw Saroun $ 149,999 inclusive:

P(B or C) = 19.9% + 24.7%
= 44.6% => 0.446

(d)

4.73 Oil Spills

(a) Occur in Ocean:

P(Atlantic) + P(Pacific) = 0.008 + 0.37
= 0.045 => 4.5%

(b) Occur in lake or harbor:

P(Great Lake) + P(Others lakes) + P(Harbors) =
0.090 + 0002 + 0.164 = 0.186 => 18.6%

(c) Do not occur in...

sum of excluded = 0.008 + 0.037 + 0.020 +
0.002 + 0.366 = 0.433

P(not in lake, o ...) = 1 - 0.433
= 0.567 (56.7%)

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4.74 Religion in America

(a) Catholic or Protestant:

$0.239 + 0.513 = 0.752$

(b) Not Jewish:

$1 - 0.017 = 0.983$

(c) Not catholic, Protestant, Jewish

$1 - (0.239 + 0.513 + 0.017)$

$= 1 - 0.769 \Rightarrow 0.231$

4.75 Ages of Senators

Use table 4.69:

(a) 50 years old or older

$P(\geq 50) = 1 - P(\text{Under } 50)$

$= 1 - \frac{12}{100} = 0.88$

(b) Under 70 years old:

$P(\text{Under } 70) = 1 - P(\geq 70)$

$= 1 - \frac{18+5}{100} \Rightarrow 0.77$

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Question 21

→ Spinner has 4 equal sectors (yellow, blue, red, green) Probability of landing red or blue.

Red sector = 1
blue sector = 1
Total favourable outcomes = 1+1 = 2

→ Mutually exclusive Events: cannot land on both red or blue at same time.

→ P (Red or blue) = P(Red) + P(blue)

→ Total outcomes = N = 4
P (Red) = 1/4
P (Blue) = 1/4

→ $P(\text{Red or Blue}) = \frac{1}{4} + \frac{1}{4} \Rightarrow \frac{2}{4} \Rightarrow \frac{1}{2} = 0.5$ 50%

Question 22

We use General Addition Rule of
P(A∪B) = P(A) + P(B) − P(A∩B)

Total Students (N) = 30
P(Girl):
no. of girls = 13

$P(\text{Girls}) = \frac{13}{30}$

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P(A Student)

Total A students = Boys + Girls with A
= 4+5 ⇒9

$$P(A\ Student) = \frac{9}{30}$$

Find P(Girl and A student):

Girls who got A = 5

$$P(\dots) = \frac{5}{30}$$

Apply General Addition Rule.

$$= P(Girls) + P(A\ student) - P(Girl\ \&\ A\ Stud)$$

$$= \frac{13}{30} + \frac{9}{30} - \frac{5}{30}$$

$$= \frac{13+9-5}{30} = \frac{17}{30} \approx 0.5667$$

$$= 56.67\%$$

Question # 27

P(king) + P(club) - P(king & club)

Total = 52
P(king) = 4/52
P(club) = 13/52
P(king of club) = 1/52

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$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{4+13-1}{52} = \frac{16}{52} = \frac{4}{13}$$

$$= 0.3077$$

$$30.77\%$$

Question #28

→ 8% defects in shaft only
→ 6% defects in bushing only
→ 2% defects in both.

* P(shafts) = 8% = 0.08
* P(bushing) = 6% = 0.06
* P(both) = 2% = 0.02

(a) P(Bushing) = P(Bushing) + P(both)
= 0.06 + 0.02 = 0.08 = 8%

(b) P(shaft or bushing) = P(shaft) + P(bushing) + P(both)
= 0.08 + 0.06 + 0.02
= 0.16 ⇒ 16%

(c) P(Exactly one) = P(shaft Only) + P(Bushing Only)
= 0.08 + 0.06 ⇒ 0.14
= 14%

(d) P(Neither) = 100% − (P(shaft or bushing)
= 100% − 16%
= 84% ⇒ 0.84

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Conditional probability:
↳ having additional info

$S = \{C_1, C_2 \dots C_{13}\}$

without additional info $\rightarrow P(C_i) = \frac{1}{13}$, $(i = 1, 2, 3 \dots 13)$

with additional info $\rightarrow P = \frac{11}{13}$ $S_R = \{C_1 \dots C_{11}\}$

$\Rightarrow$ Two dice rolled:

$n(S) = 36$

$P(A) = 5/36$

$n(S_R) = 30$

$P(A) = 4/30$

$\rightarrow$ A and K in MTH262, Prove that K will
$\Rightarrow$ join musical night, while A is already at venue since last 7 days.

$$P(K|A) = \frac{P(K \cap A)}{P(A)} = \frac{n(K \cap A)/n(S)}{n(A)/n(S)}$$

$$= \frac{n(K \cap A)}{n(A)}$$

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Exercise 4.4

find that probability that order was filed by:

(a) A black

(b) A white femal

(c) A male, given that filter was white.

(d) A male, given that filter was black.

(a) $P(R_2) = \frac{\text{Total Black}}{\text{Total}} = \frac{147}{407} = 0.3611$

(b) white female:

$P(R_1 \text{ and } G_2) = \frac{210}{407} \Rightarrow 0.5159$

(c) A male, given that filter was white.

$P(\text{male} | \text{white})$

$P(G_1 | R_1) = \frac{P(G_1 \text{ and } R_1)}{P(R_1)}$

$\frac{P(G_1 \cap R_1)/n(S)}{P(R_1)/n(S)} = \frac{30/407}{240/407} = \frac{30}{240}$

$= 0.125$

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(d)

$$P(G_1 | R_2) = \frac{P(G_1 \cap R_2)/n(S)}{P(R_2)/n(S)}$$

$$= \frac{26/407}{147/407} = \frac{26}{147} = 0.178$$

Question # 37

P (passed both) = $P(F1 \cap F2)$ = 25% = 0.25

P (passed first) = $P(F1)$ = 42% = 0.42

To find : P (passed second | passed first)

$$P(F2 | F1) = \frac{P(F1 \cap F2)}{P(F1)}$$

$$= \frac{0.25}{0.42} = 0.5952 \Rightarrow 59.52\%$$

Question # 38

P (Black 1st & white 2nd) = $P(B \cap W)$

P (Black on 1st) = $P(B)$ = 0.47

To find : P (white on 2nd | Black on 1st)

$$P(W2 | B1) = \frac{P(B1 \cap W2)}{P(B1)} = \frac{0.34}{0.47}$$

$$= 0.7234 \Rightarrow 72.34\%$$

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without replacement ⇒ select object → observe → no return.

with replacement ⇒ select object → observe → then return.

Dependent Event ⇒ results changes due to other event.

Independent Event ⇒ results not changes bez of other event.

(A > K > D)